

Artificial intelligence

PROBLEM SOLVING AGENTS

Problem-solving agents are a type of goal-based agent that use atomic representations, where states of the world are treated as complete wholes without any internal structure. Goal-based agents that employ more advanced factored or structured representations are generally referred to as planning agents.

Goals are essential as they organize behaviour by defining the objectives the agent aims to achieve, thereby limiting the actions it needs to consider.

Steps Required to Solve a Problem

The process typically involves two main steps:

1.Goal Formulation

This step involves defining the goals that require some action to be achieved, based on the agent's current situation and its performance measure. The goal is defined as a set of world states in which the goal is satisfied. Reaching from the initial state to the goal state requires actions that cause transitions between world states. Actions should be abstract enough to simplify decision-making, such as “turn left” instead of “turn left 30 degrees,” to reduce uncertainty and complexity.

2.Problem Formulation

This process involves deciding what actions and states to consider in order to achieve the formulated goal.

For example, when driving from Ernakulam to Chennai:

States: could represent major towns or places in between.

Actions: could be abstract operations like “turn left,” “turn right,” “go straight,” or even higher-level actions like driving from one major town to another.

3.Well-Defined Problems and Solutions

A problem can be formally defined by the following five components:

1. Initial state: The state where the agent starts.
2. Possible actions: The set of actions available to the agent in a given state.
3. Transition model: Describes what each action does, represented by a function that returns the state resulting from performing an action in a given state.

A successor is any state reachable from another by a single action.

The state space is the set of all states reachable from the initial state, forming a directed graph where nodes represent states and links represent actions.

A path is a sequence of states connected by actions.

4. Goal test: Determines whether a given state is a goal state.
5. Path cost function: Assigns a numeric cost to each path, usually the sum of the costs of the actions along the path.

A solution is an action sequence that leads from the initial state to the goal state. An optimal solution is one that has the lowest path cost among all possible solutions.

4. Abstraction

Abstraction is the process of removing irrelevant details and retaining only essential information about the states. It simplifies problem-solving by focusing on the most important aspects needed for searching. Choosing a good abstraction involves removing unnecessary details while maintaining validity and ensuring that abstract actions remain easy to execute. The ability to create useful abstractions is crucial, as it helps intelligent agents manage the complexity of the real world.

MCQ QUESTIONS

1. Which of the following best defines a problem-solving agent?

- A. An agent that acts only on current precepts
 - B. An agent that finds a sequence of actions to achieve a goal
 - C. An agent that learns from past experiences only
 - D. An agent that reacts randomly to stimuli
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2. In a problem-solving process, which is the correct order of steps?

- A. Problem formulation → Goal formulation → Search → Execution
 - B. Goal formulation → Problem formulation → Search → Execution
 - C. Execution → Search → Problem formulation → Goal formulation
 - D. Problem formulation → Search → Execution → Goal formulation
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3. What is the 'transition model' in a problem-solving agent?

- A. It defines the set of actions available to the agent
 - B. It defines how the world changes after an action is performed
 - C. It defines the final goal state
 - D. It defines the cost of performing an action
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4. Which of the following is NOT a component of a well-defined problem?

- A. Initial state
 - B. Actions
 - C. Transition model
 - D. Agent program
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5. Which type of agent uses search and planning to achieve its goals?

- A. Simple reflex agent

- B. Model-based reflex agent
 - C. Goal-based agent
 - D. Utility-based agent
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6. Which of the following search types uses heuristics to guide the search?

- A. Depth-First Search
 - B. Breadth-First Search
 - C. Uniform Cost Search
 - D. A* Search
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7. Which agent is best suited when multiple goals with different desirability levels exist?

- A. Goal-based agent
 - B. Utility-based agent
 - C. Reflex agent
 - D. Learning agent
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8. What is the main advantage of a learning agent?

- A. It performs only reflexive actions
 - B. It improves performance based on experience
 - C. It ignores feedback from the environment
 - D. It cannot adapt to new situations
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9. Which of the following environments best suits a problem-solving agent?

- A. Stochastic and dynamic
 - B. Deterministic and fully observable
 - C. Partially observable and random
 - D. Competitive and uncertain
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10. In a route-finding problem, what does the path cost usually represent?

- A. The number of possible routes
 - B. The total distance or energy used
 - C. The number of actions available
 - D. The initial state value
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11. Which of the following is an example of a problem-solving agent application?

- A. Automatic room lighting system
 - B. GPS navigation system
 - C. Thermostat temperature control
 - D. Simple spam filter
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12. In AI problem-solving, the goal test is used to:

- A. Measure the path cost of all actions
 - B. Determine whether a given state satisfies the goal condition
 - C. Predict the next possible state
 - D. Select the best heuristic function
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13. Which of the following search algorithms is considered uninformed?

- A. A* Search
 - B. Greedy Best-First Search
 - C. Depth-First Search
 - D. Hill Climbing
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14. The path cost function is primarily used to:

- A. Track the number of visited nodes

- B. Evaluate the total cost of a solution path
 - C. Measure the complexity of a problem
 - D. Count the number of available actions
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15. Which type of agent maintains an internal model of the world's current state?

- A. Simple reflex agent
- B. Model-based reflex agent
- C. Goal-based agent
- D. Utility-based agent

Provided link for Problem Solving in Artificial Intelligence is:

<https://www.youtube.com/watch?v=sKmNC-kvlfY>